Mass and Stiffness matrices for P1 finite elements over surface meshes

Let ABC be a triangle given by three vertices $A, B, C \in \mathbb{R}^3$.

We have seen already that the building blocks for the mass and stiffness matrices are the quantities

$$\int_{ABC} \varphi_P \varphi_Q \qquad \forall P, Q \in \{A, B, C\}$$

for the mass, and

$$\int_{ABC} \nabla\varphi_P \cdot \nabla\varphi_Q \qquad \forall P, Q \in \{A, B, C\}$$

for the stiffness, where, $\forall P \in \{A, B, C\}$, φ_P refers to the (unique) function which is affine (on the affine space containing ABC) and has value 1 at P and value 0 at the other two vertices of ABC.

Although the functions $\varphi_A, \varphi_B, \varphi_C$ are very simple, the above integrals requires a bit of work because there is no canonical reference frame in which to express them.

To circumvent this difficulty, we consider a reference triangle $\hat{A}\hat{B}\hat{C}$ in $\mathbb{R}^2$ (note: 2, not 3) where $\hat{A} = (0,0)$, $\hat{B} = (1,0)$, $\hat{C} = (0,1)$ and we consider the map

$$\begin{aligned} \Phi : \mathbb{R}^2 &\longrightarrow \mathbb{R}^3 \\ (s,t) &\longmapsto A + s(B-A) + t(C-A) \end{aligned}$$

which is affine, and a diffeomorphism between $\hat{A}\hat{B}\hat{C}$ and ABC (viewed as a 2D submanifold of $\mathbb{R}^3$) provide ABC is not degenerate (i.e. A, B, C are not aligned).

This amounts to choosing a basis $(B-A, C-A)$ and an origin A in the plane containing ABC, to view it as a 2D vector space.

That basis (in general) is not orthonormal, for the scalar product induced by the one in $\mathbb{R}^3$, and more importantly the surface measure (in math terms, the volume form) is NOT $ds\,dt$!

By the change of variable formula,

$$\int_{ABC} \varphi_P \varphi_Q \, d\mathrm{surf} = \int_{\Phi(\hat{A}\hat{B}\hat{C})} \varphi_P \varphi_Q \, d\mathrm{surf}$$

$$= \int_{\hat{A}\hat{B}\hat{C}} (\varphi_P \circ \Phi)(\varphi_Q \circ \Phi) \sqrt{\det(d\Phi^t d\Phi)} \, ds\,dt,$$

i.e. the pull-back of the surface measure on ABC by Φ is the measure

$$\sqrt{\det(d\Phi^t d\Phi)} \, ds\,dt.$$

<u>Rmk</u> 1) $d\Phi \in \mathcal{L}(\mathbb{R}^2, \mathbb{R}^3)$ is represented by a matrix in $\mathbb{R}^{3,2}$

so $d\Phi^t d\Phi \in \mathbb{R}^{2,3} \times \mathbb{R}^{3,2} = \mathbb{R}^{2,2}$

is square and has a well defined determinant.

When $\Phi : \mathbb{R}^n \to \mathbb{R}^n$ (a version of the change of variable formula you might be more familiar with) $d\Phi \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^n)$ has itself a well defined determinant and

$$\sqrt{\det(d\Phi^t d\Phi)} = \sqrt{\det(d\Phi^t)\det(d\Phi)}$$

$$= \sqrt{\det(d\Phi)^2} = |\det(d\Phi)|$$

$$= |\mathrm{Jac}\,\Phi|$$

In our case above where Φ maps into a submanifold of a larger space, only the first expression $\sqrt{\det(d\Phi^t d\Phi)}$ makes sense.

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2) Since Φ and φ_P, $\varphi_{\hat{P}}$ are all affine, it is immediate that

equivalent of φ_P but on $\hat{A}\hat{B}\hat{C}$

$$\varphi_{\hat{P}} = \varphi_P \circ \Phi$$

(this wouldn't hold for P_2 elements with quadratic functions)

and therefore

$$\int_{ABC} \varphi_P \varphi_Q \, d\mathrm{surf} = \int_{\hat{A}\hat{B}\hat{C}} \varphi_{\hat{P}} \varphi_{\hat{Q}} \sqrt{\det(d\Phi^t d\Phi)} \, ds\,dt.$$

Besides, $\sqrt{\det(d\Phi^t d\Phi)}$ is a constant function (because $d\Phi$ is constant if Φ is affine).

Using the change of variable formula for a constant integrand it immediately follows that

$$\sqrt{\det(d\Phi^t d\Phi)} = \frac{\mathrm{Area}(ABC)}{\mathrm{Area}(\hat{A}\hat{B}\hat{C})},$$

but we can also check it by an elementary computation :

$$d\Phi = \left(B-A \,\middle|\, C-A \right) \in \mathbb{R}^{3,2}$$

so

$$d\Phi^t d\Phi = \left(\frac{B-A}{C-A} \right) \left(B-A \,\middle|\, C-A \right)$$

$$= \begin{pmatrix} \|B-A\|^2 & (B-A)\cdot(C-A) \\ (B-A)\cdot(C-A) & \|C-A\|^2 \end{pmatrix}$$

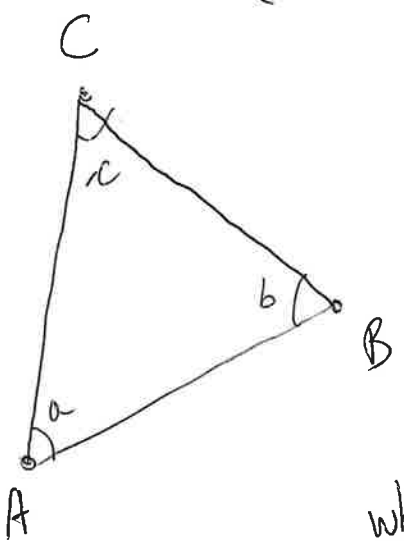
hence

$$\det\left(d\Phi^t d\Phi\right) = \|B-A\|^2 \|C-A\|^2 - \left((B-A)\cdot(C-A)\right)^2$$

$$= \|B-A\|^2 \|C-A\|^2 - \left(\|B-A\| \, \|C-A\| \cos a\right)^2$$

$$= \|B-A\|^2 \|C-A\|^2 (1 - \cos^2 a)$$

$$= \left(\|B-A\| \cdot \|C-A\| \sin a\right)^2$$



where we call "a" the angle at A.

$$= \left(2\,\mathrm{Area}(ABC)\right)^2$$

Thus

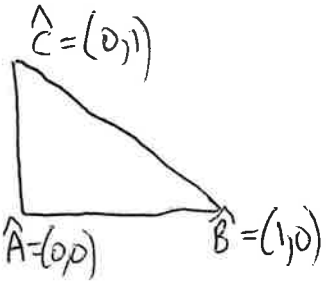
$$\sqrt{\det\left(d\Phi^T d\Phi\right)} = 2\,\mathrm{Area}(ABC) = \frac{\mathrm{Area}(ABC)}{\mathrm{Area}(\hat{A}\hat{B}\hat{C})}$$

$$\left(\text{since } \mathrm{Area}(\hat{A}\hat{B}\hat{C}) = 1/2\right),$$

and we have

$$\boxed{\int_{ABC} \varphi_P \varphi_Q = 2\,\mathrm{Area}(ABC) \int_{\hat{A}\hat{B}\hat{C}} \varphi_{\hat{P}} \varphi_{\hat{Q}}}$$

The integrals on the right-hand side are now straightforward to compute:



Using that
$$\varphi_{\hat{A}}(s,t) = 1-(s+t)$$
$$\varphi_{\hat{B}}(s,t) = s$$
$$\varphi_{\hat{C}}(s,t) = t$$

we obtain

$$\int_{\hat{A}\hat{B}\hat{C}} \varphi_{\hat{P}} \varphi_{\hat{Q}} \overset{\text{Fubini}}{=} \int_0^1 \int_0^{1-s} \varphi_{\hat{P}}(s,t)\, \varphi_{\hat{Q}}(s,t)\, dt\, ds$$

$$= \begin{cases} 1/12 & \text{if } \hat{P} = \hat{Q} \\ 1/24 & \text{if } \hat{P} \neq \hat{Q} \end{cases}$$

In summary

$$M_{ABC} = \text{Area}(ABC) \begin{pmatrix} 1/6 & 1/12 & 1/12 \\ 1/12 & 1/6 & 1/12 \\ 1/12 & 1/12 & 1/6 \end{pmatrix}$$

In practice that matrix M_{ABC} is never stored anywhere, its coefficients are inserted (added) in the correct position in the overall mesh mass matrix (of sparse class), and therefore only Area(ABC) needs to be computed, and $1/6$ an $1/12$ stored :

recall that an indexed triangular mesh is given by

- A set of vertices $V[i]$ $1 \leq i \leq nvert$ where each $V[i] \in \mathbb{R}^3$
- A set of triangle indices $T[k]$ $1 \leq k \leq ntri$ where each $T[k] \in \mathbb{N}^3$ (even: $T[k] \in \{1, \dots, nvert\}^3$) and if $T[k] = (n_1, n_2, n_3)$ the corresponding "physical" triangle is $(V[n_1], V[n_2], V[n_3])$

To <u>assemble</u> the mass matrix of the mesh,

this verb is commonly used to denote that process

we therefore do:

1) Initialize a sparse matrix $M \in \mathbb{R}^{nvert \times nvert}$ with suitable sparse structure (i.e $M[i,j]$ as a reserved storage only when $V[i]$ and $V[j]$ are neighbors in the mesh, i.e. part of a common triangle
2) Set all coefficients (of course only the reserved ones!) to zero.

3) Fill M : in pseudo code

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for k = 1 ... ntri :
    let (n1, n2, n3) = T[k]
    let A = V[n1] , B = V[n2], C = V[n3]
    compute S = area(A, B, C)
    M[n1, n1] += S/6 ;
    M[n2, n2] += S/6 ;
    M[n3, n3] += S/6 ;
    M[n1, n2] += S/12 ;
    M[n1, n3] += S/12 ;
    M[n2, n3] += S/12 ;
    etc for the 3 symmetric remaining ones.
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The implementation might use the symmetry of M (to avoid to store one half of the off diagonal coefficients

How you get access to M[i, j] also depends on the choice of implementation for the sparse matrix M (COO, CSR, CSC, hash matrix, ...)

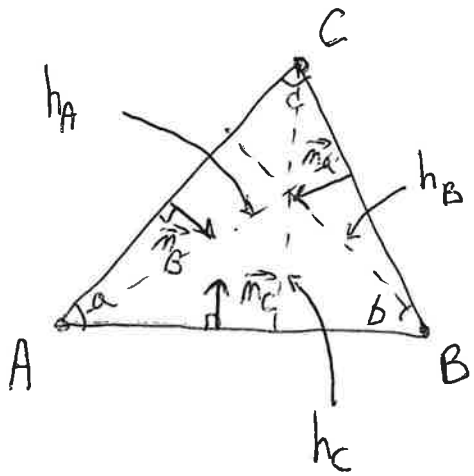
We now turn to the stiffness matrix, again by considering the building block of one triangle ABC.

We shall present two approaches. The first one is "pedestrian" and conceptually simpler, it doesn't rely on a change of variable formula but is also less generalizable to more complex situations.

<u>Approach 1</u> Because the φ_P and φ_Q are affine functions, their gradients $\nabla\varphi_P$, $\nabla\varphi_Q$ are constant functions (constant vectors) and therefore

$$\int_{ABC} \nabla\varphi_P \cdot \nabla\varphi_Q \, d\mathrm{surf} = \mathrm{Area}(ABC) \underbrace{\nabla\varphi_P \cdot \nabla\varphi_Q}_{c^{st} \in \mathbb{R}} .$$

To compute $\nabla\varphi_P \cdot \nabla\varphi_Q$ lets us introduce a few notations over the triangle



h_P = height at vertex P

$\vec{n}_P$ = inward unit normal to the edge opposing P

$\uparrow$ = angle at vertex P

We have

$$\nabla \varphi_P = \frac{1}{h_P} \vec{n}_P \qquad \forall P \in \{A, B, C\}$$

and (we provide the interaction $B \leftrightarrow C$, the others are perfectly identical after relabelling)

$$h_B = AB \sin(a) \qquad \left(\text{here } AB := \|B-A\|, \; CA := \|A-C\|\right)$$

$$h_C = CA \sin(a)$$

$$\vec{n}_B \cdot \vec{n}_C = -\cos(a)$$

$$2 \text{ Area}(ABC) = AB \times CA \times \sin(a)$$

$$(A-C) \cdot (B-A) = -CA \times AB \times \cos(a)$$

hence

$$\int_{ABC} \nabla \varphi_B \nabla \varphi_C = -\frac{1}{2} \operatorname{cotan}(a) = \frac{(A-C)\cdot(B-A)}{4 \text{ Area}(ABC)}$$

better for human understanding

better for computer evaluation

Similarly (no need to recompute, just circular permutation of ABC)

$$\int_{ABC} \nabla \varphi_C \nabla \varphi_A = -\frac{1}{2} \operatorname{cotan}(b) = \frac{(B-A)\cdot(C-B)}{4 \text{ Area}(ABC)}$$

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$$\int_{ABC} \nabla\varphi_A \nabla\varphi_B = -\frac{1}{2}\cot(c) = \frac{(A-C)\cdot(C-B)}{4\,\text{Area}(ABC)}$$

Regarding the self-interactions (i.e. diagonal of S)

stiffness matrix

$$\nabla\varphi_A = \frac{1}{AB\,\sin(b)}\,\vec{n}_A$$

because $h_A = \begin{cases} AB\sin(b) \\ CA\sin(c) \end{cases}$

depending how we look at it

$$= \frac{1}{CA\,\sin(c)}\,\vec{n}_A$$

so

$$\int_{ABC} \nabla\varphi_A \cdot \nabla\varphi_A = \frac{\text{Area}(ABC)}{AB \times CA \times \sin(b)\sin(c)} = \frac{(BC)^2}{4\,\text{Area}(ABC)}$$

good for evaluation by computer

and similarly

$$\int_{ABC} |\nabla\varphi_B|^2 = \frac{(CA)^2}{4\,\text{Area}(ABC)}, \quad \int_{ABC} |\nabla\varphi_C|^2 = \frac{(AB)^2}{4\,\text{Area}(ABC)}$$

Using the angle-side-angle formula for the area, i.e.

$$\text{Area}(ABC) = \frac{(AB)^2\sin(a)\sin(b)}{2\sin(c)} = \frac{(BC)^2\sin b\,\sin(c)}{2\sin(a)} = \frac{(CA)^2\sin(c)\sin(a)}{2\sin(b)}$$

we get $\int_{ABC} |\nabla\varphi_A|^2 = \frac{\sin(a)}{2\sin(b)\sin(c)}$; $\int_{ABC} |\nabla\varphi_B|^2 = \frac{\sin(b)}{2\sin(a)\sin(c)}$; $\int_{ABC} |\nabla\varphi_C|^2 = \frac{\sin(c)}{2\sin(a)\sin(b)}$

Approach 2

Here we rely instead on the change of variable formula:

$$(*) \qquad \int_{ABC = \Phi(\hat{A}\hat{B}\hat{C})} \nabla\varphi_P \cdot \nabla\varphi_Q = \int_{\hat{A}\hat{B}\hat{C}} (\nabla\varphi_P \circ \Phi)\cdot(\nabla\varphi_Q \circ \Phi) \sqrt{\det(d\Phi^t \circ d\Phi)}$$

As already observed for the computation of the mass matrix,

$$\varphi_{\hat{P}} = \varphi_P \circ \Phi ,$$

hence by the rule of differentiation of composite functions (i.e. chain rule)

$$(**) \qquad \nabla\varphi_{\hat{P}} = (\nabla\varphi_P \circ \Phi) \circ d\Phi$$

This does not immediately allow to rewrite the integrand expression $(\nabla\varphi_P \circ \Phi)\cdot(\nabla\varphi_Q \circ \Phi)$ in terms of $\nabla\varphi_{\hat{P}}$ and $\nabla\varphi_{\hat{Q}}$, because of the trailing "$\circ d\Phi$".

If Φ was a diffeo between (some open of) $\mathbb{R}^n$ to (some other open) of the same $\mathbb{R}^n$, we could write

$$\nabla\varphi_P \circ \Phi = \nabla\varphi_{\hat{P}} \circ (d\Phi)^{-1} ,$$

and then substitute into (*). But this cannot be done here since $d\Phi \in \mathcal{L}(\mathbb{R}^2, \mathbb{R}^3)$ is not square. MS15

Instead for a (symmetric) 2x2 matrix G (we call it G since it shall play the role of a Riemannian metric), we write from (**)

$$\nabla\varphi_{\hat{P}}\, G\, \nabla\varphi_{\hat{Q}}^{\,t} = (\nabla\varphi_P \circ \Phi)\underbrace{d\Phi\, G\, d\Phi^t}_{\in \mathbb{R}^{3\times 2} \cdot \mathbb{R}^{2,2} \cdot \mathbb{R}^{2,3} = \mathbb{R}^{3,3}}(\nabla\varphi_Q \circ \Phi)^t$$

Ideally, in order to reproduce $(\nabla\varphi_P \circ \Phi)\cdot(\nabla\varphi_Q \circ \Phi)$ we would want $d\Phi G d\Phi^t = I_{3\times 3}$. This is not possible since $d\Phi G d\Phi^t$ is at most of rank 2.

Yet, if we choose $G = (d\Phi^t d\Phi)^{-1}$,

then

$$d\Phi G d\Phi^t = d\Phi(d\Phi^t d\Phi)^{-1} d\Phi^t$$

is a projection onto the tangent plane (here simply the plane containing the triangle ABC, or more precisely parallel to it and containing the origin, since it must be a vector space) of the surface parametrize by Φ, at the corresponding point. Indeed, remember that a projection is a linear map U satisfying $U^2 = U$.

Here if $U := d\Phi\,(d\Phi^t d\Phi)^{-1} d\Phi^t$, then

$$U^2 = d\Phi \underbrace{(d\Phi^t d\Phi)^{-1} d\Phi^t d\Phi}_{= I_{2\times 2}} (d\Phi^t d\Phi)^{-1} d\Phi^t$$

$$= U,$$

and clearly $\mathrm{Im}(U) = \mathrm{Im}(d\phi)$.

Since $\nabla\varphi_P$ and $\nabla\varphi_Q$ both belong to $\mathrm{Im}(d\phi)$ (the gradients live in the tangent plane)

matrix product

$$(\nabla\varphi_P \circ \Phi)\, U\, (\nabla\varphi_Q \circ \Phi)^t = (\nabla\varphi_P \circ \Phi)(\nabla\varphi_Q \circ \Phi)^t$$
$$= (\nabla\varphi_P \circ \Phi) \cdot (\nabla\varphi_Q \circ \Phi)$$

dot product

Hence :

$$\int_{ABC} \nabla\varphi_P \cdot \nabla\varphi_Q = \int_{\hat{A}\hat{B}\hat{C}} \nabla\varphi_{\hat{P}}\, G\, \nabla\varphi_{\hat{Q}} \sqrt{\det(G^{-1})}$$

Let us now compute more explicitely G and G^{-1} when $\Phi(s,t) = A + s(B-A) + t(C-A)$.

We have

$$d\Phi = \left(B-A \,\middle|\, C-A \right)$$

$$d\Phi^t = \begin{pmatrix} B-A \\ \hline C-A \end{pmatrix}$$

- $$G^{-1} = d\Phi^t d\Phi = \begin{pmatrix} \|B-A\|^2 & (B-A)\cdot(C-A) \\ (B-A)\cdot(C-A) & \|C-A\|^2 \end{pmatrix}$$

- $$\det(G^{-1}) = \|B-A\|^2\|C-A\|^2 - \|B-A\|^2\|C-A\|^2 \sin^2 a$$
$$= \left(\|B-A\|\,\|C-A\| \sin a\right)^2 = 4\,\text{Area}(ABC)^2$$

- $$G = \frac{1}{\det(G^{-1})} \begin{pmatrix} \|C-A\|^2 & -(B-A)(C-A) \\ -(B-A)(C-A) & \|B-A\|^2 \end{pmatrix}$$

$$\nabla\varphi_{\hat{A}} = (-1,-1), \quad \nabla\varphi_{\hat{B}} = (1,0), \quad \nabla\varphi_{\hat{C}} = (0,1)$$

and plugging these into

$$\int_{ABC} \nabla\varphi_{\hat{P}}\, G\, \nabla\varphi_{\hat{Q}} \sqrt{\det(G^{-1})} = \text{Area}(ABC)\, \nabla\varphi_{\hat{P}}\, G\, \nabla\varphi_{\hat{Q}} \sqrt{\det(G^{-1})}$$

we (should) recover the exact same values as the ones obtained the "pedestrian" way. This second approach is more likely to be extended to, say, curved elements.

The construct of the assembled stiffness matrix follows then the exact same lines as those described in pages MS9 - MS10.

As a matter of fact, since the computations involved are also very similar, it might be worth performing both in a same step to avoid doubling computations (this optimization has not been done in the proposed header files on github).